

Wireless and Cellular Communications (EE5141)

Simulation Assignment 1

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1 Question 1

1.1 Background

Jakes' Doppler spectrum takes the form:

$$S_h(f) = \begin{cases} \frac{1}{\pi f_d \sqrt{1 - (f/f_d)^2}}, & |f| \leq f_d \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

The corresponding autocorrelation function is:

$$R_h(\tau) = J_0(2\pi f_d \tau) \quad (2)$$

where J_0 denotes the zeroth-order Bessel function of the first kind.

Considering sampling time T_s , the discrete channel autocorrelation becomes:

$$R[m] = J_0(2\pi f_d m T_s) \quad (3)$$

1.2 A) Levinson Algorithm

Theory

We want to design a linear noise-colouring filter that converts white Gaussian noise into a process with Jakes' spectrum.

We seek a sequence $h[k]$ such that:

$$\mathbb{E}[h[k] h^*[k+m]] = R[m] = J_0(2\pi f_d m T_s) \quad (4)$$

Starting with $w[k] \sim \mathcal{CN}(0, 1)$ (i.i.d.), we use an auto-regressive (AR) model:

$$h[k] = \sum_{i=1}^p a_i h[k-i] + w[k] \quad (5)$$

We must choose a_i such that the autocorrelation equals $R[m]$. Taking expectations:

$$\begin{aligned} \mathbb{E}[h[k] h^*[k-m]] &= \sum_{i=1}^p a_i \mathbb{E}[h[k-i] h^*[k-m]] + \mathbb{E}[w[k] h^*[k-m]] \\ R[m] &= \sum_{i=1}^p a_i R[m-i] \end{aligned} \quad (6)$$

The last equality holds because $h[k-m]$ depends only on past noise values. This system can be solved efficiently using the *Levinson-Durbin recursion*.

After obtaining the coefficients a_i , white noise $w[k]$ is passed through the AR filter (5) to produce $h[k]$.

Results

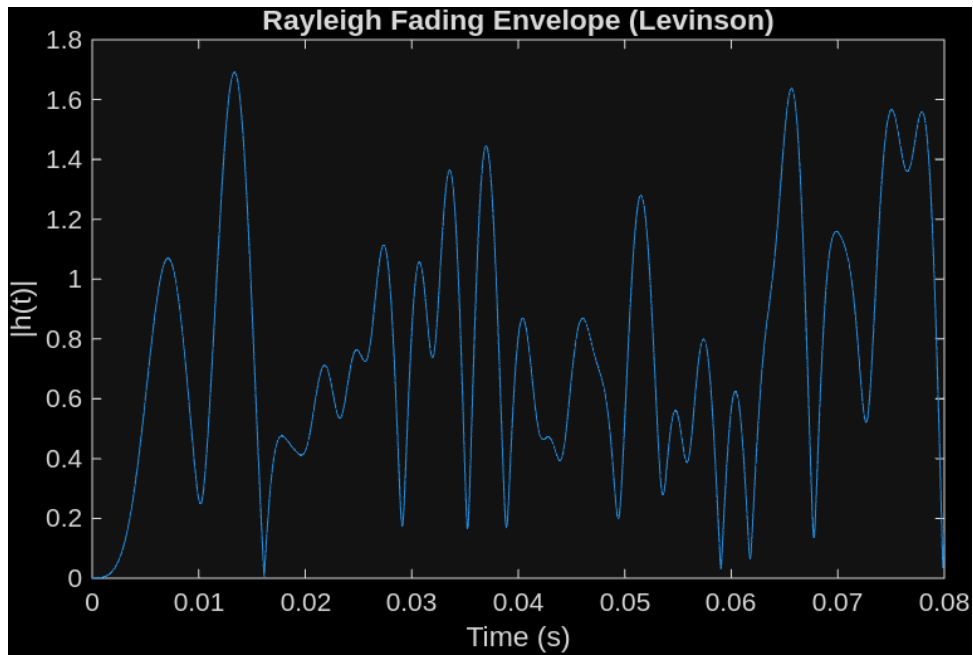


Figure 1: Path gain magnitude (Levinson method).

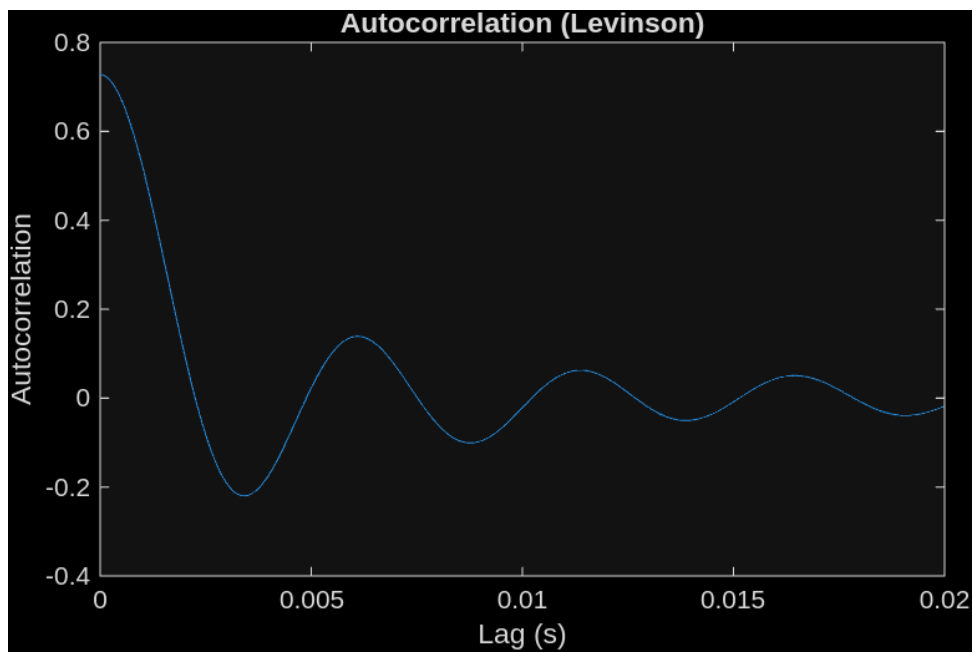


Figure 2: Autocorrelation (Levinson method) — closely matches the zeroth-order Bessel function.

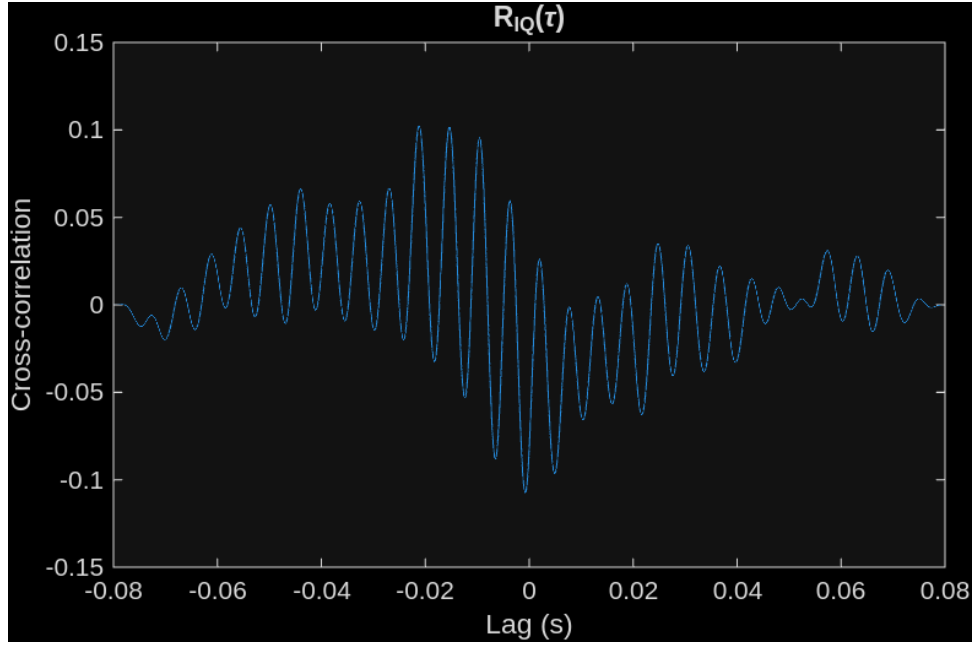


Figure 3: Cross-correlation between real and imaginary parts (Levinson method) — low (≈ 0.1), confirming near-independence.

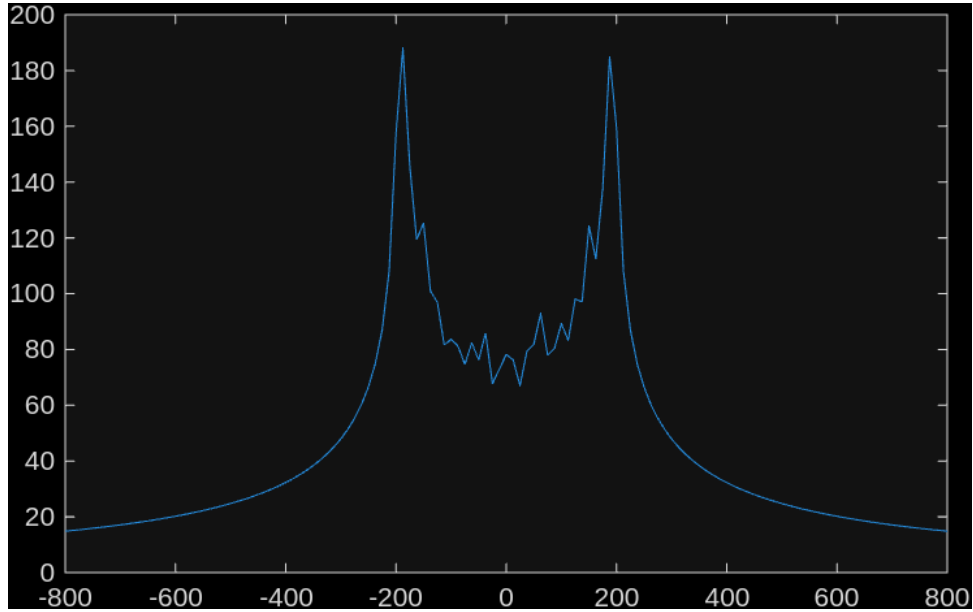


Figure 4: Power spectral density (Levinson method) — follows Jakes' spectrum.

1.3 B) Smith's Model

Theory

1. Generate complex white Gaussian noise in the frequency domain:

$$W(f_k) = W_I(f_k) + j W_Q(f_k)$$

2. Multiply by the shaping filter $\sqrt{S_h(f_k)}$:

$$H(f_k) = W(f_k) \cdot \sqrt{S_h(f_k)}$$

3. Take the IFFT to obtain the time-domain fading:

$$h[n] = \text{IFFT}\left\{\tilde{H}(f_k)\right\}$$

Results

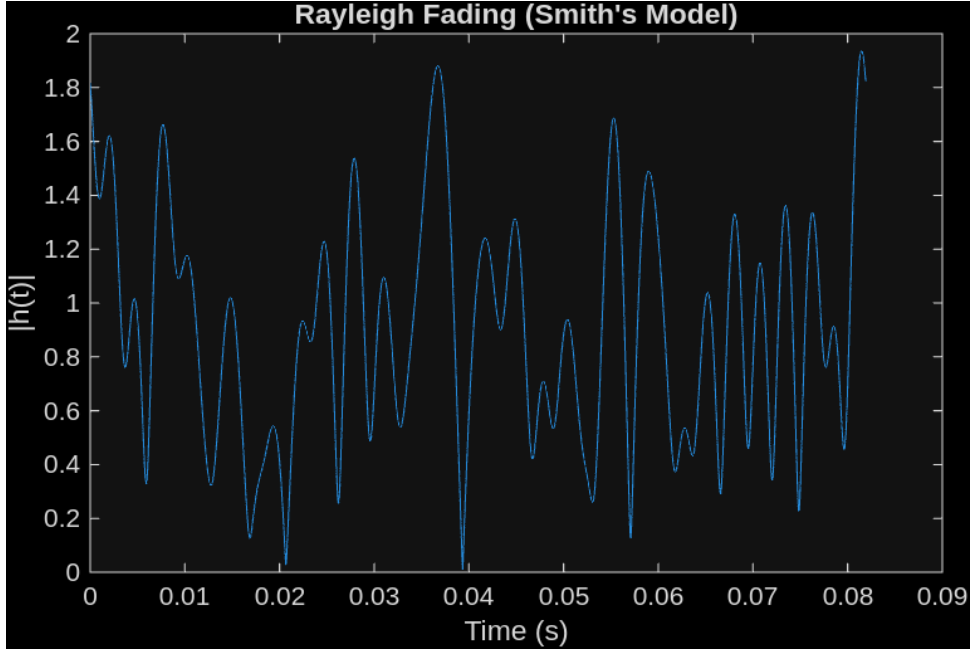


Figure 5: Path gain magnitude (Smith's model).

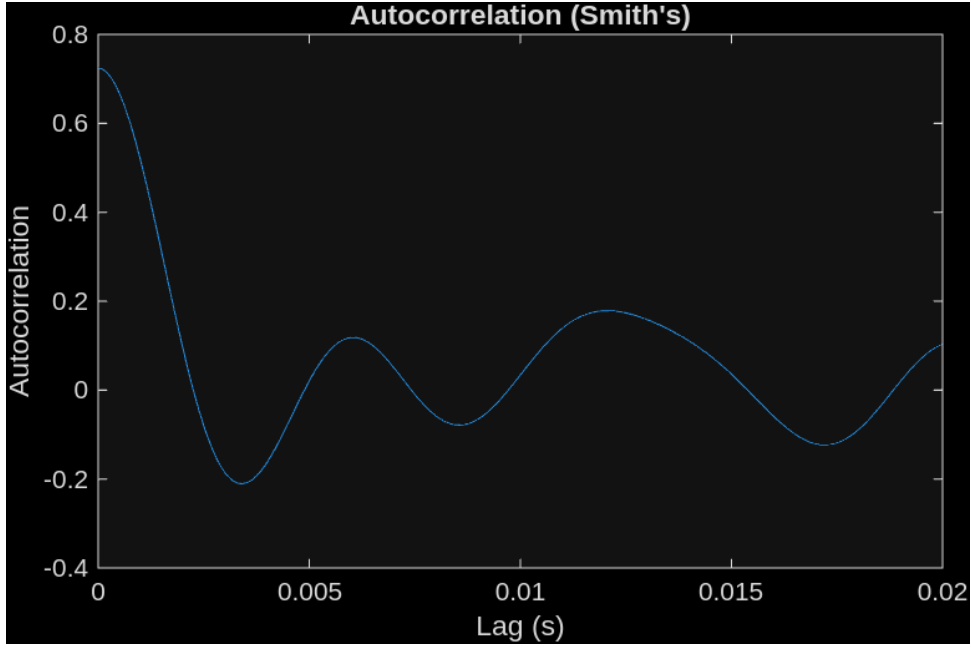


Figure 6: Autocorrelation (Smith's model).

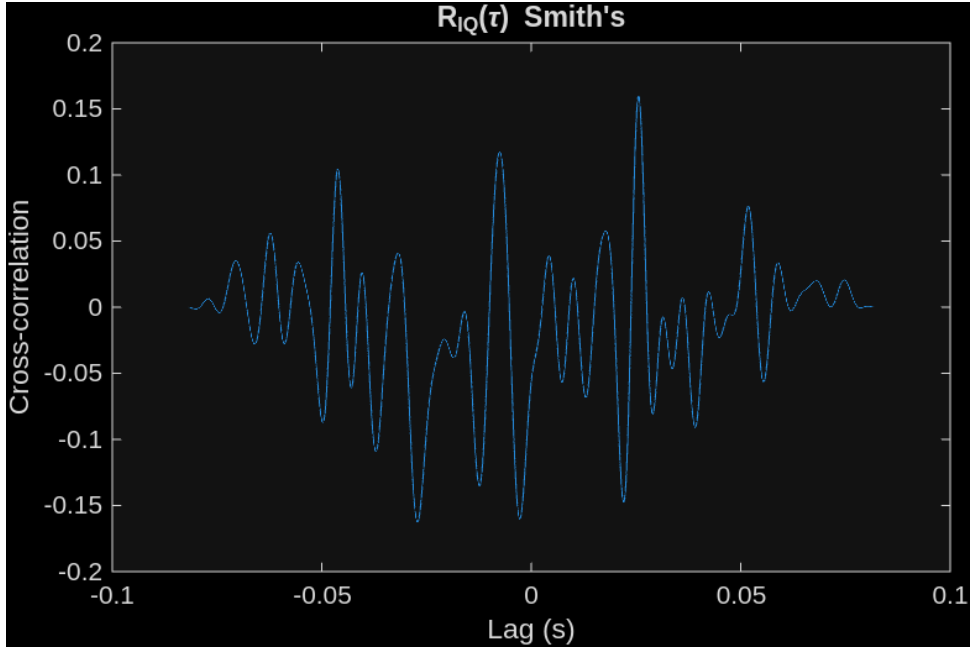


Figure 7: Cross-correlation (Smith's model).

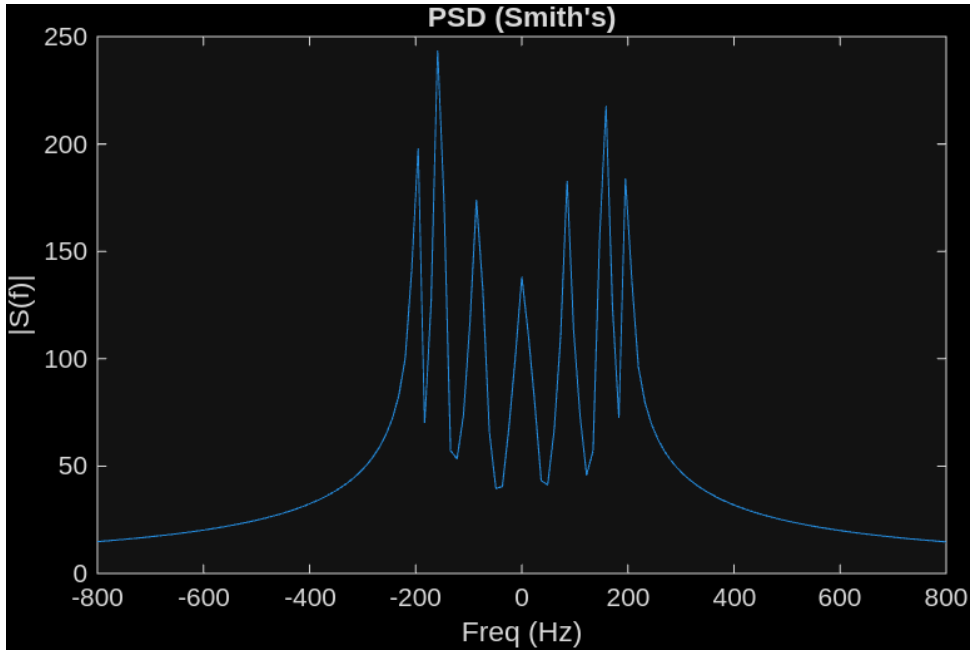


Figure 8: Power spectral density (Smith's model).

1.4 C) Modified Sum of Sinusoids

Theory

Clarke's model represents the channel as:

$$h(t) = \sum_{n=1}^{N_0} a_n e^{j(2\pi f_n t + \phi_n)} \quad (7)$$

In the modified sum-of-sinusoids (SOS) model, this is approximated by:

$$h_I(t) = \sum_{n=1}^N \cos(2\pi f_n t + \phi_n) \quad (8)$$

$$h_Q(t) = \sum_{n=1}^N \sin(2\pi f_n t + \psi_n) \quad (9)$$

The composite complex fading envelope is:

$$h(t) = \sqrt{\frac{2}{N}} \sum_{n=1}^N [\cos(2\pi f_n t + \phi_n) + j \sin(2\pi f_n t + \psi_n)] \quad (10)$$

where the Doppler frequency for the n -th component is:

$$f_n = f_d \cos(\theta_n) \quad (11)$$

The parameters are distributed as:

$$\phi_n, \psi_n \sim \mathcal{U}(0, 2\pi), \quad \theta_n = \frac{\pi(n - 0.5)}{N} \quad (12)$$

Results

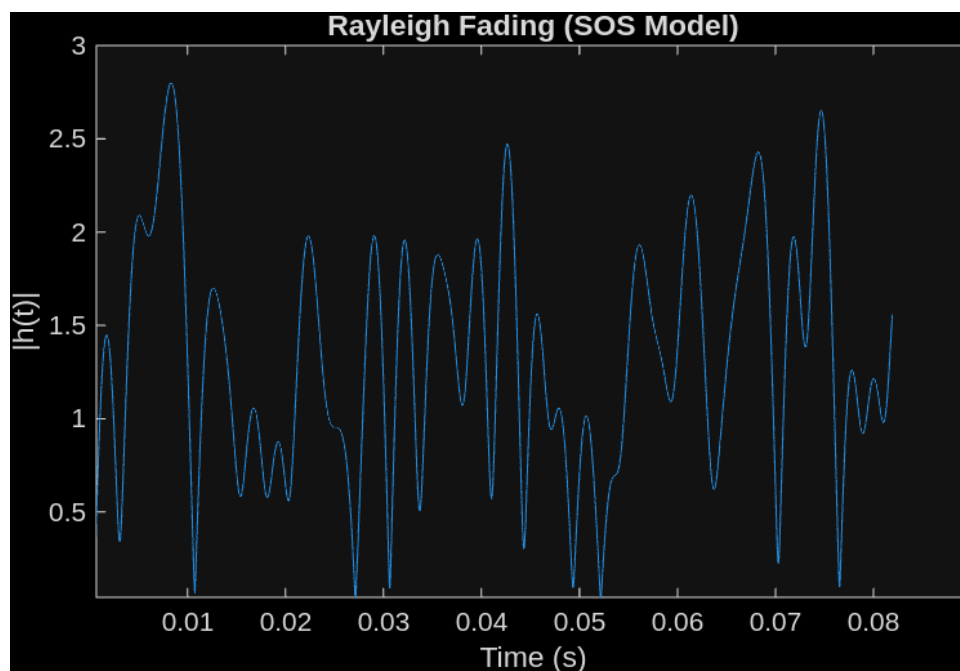


Figure 9: Path gain magnitude (SOS model).

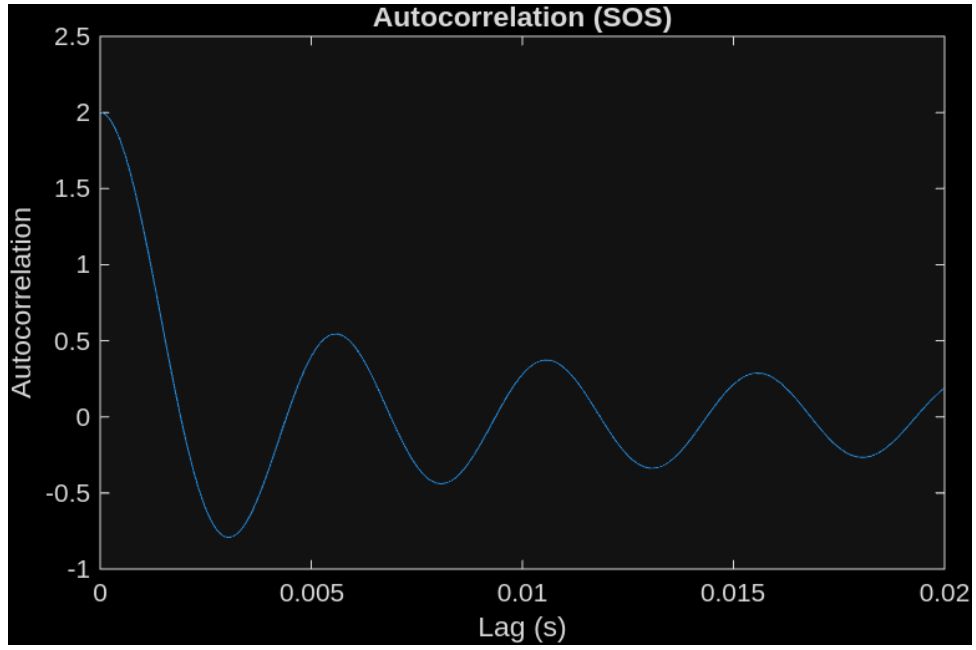


Figure 10: Autocorrelation (SOS model).

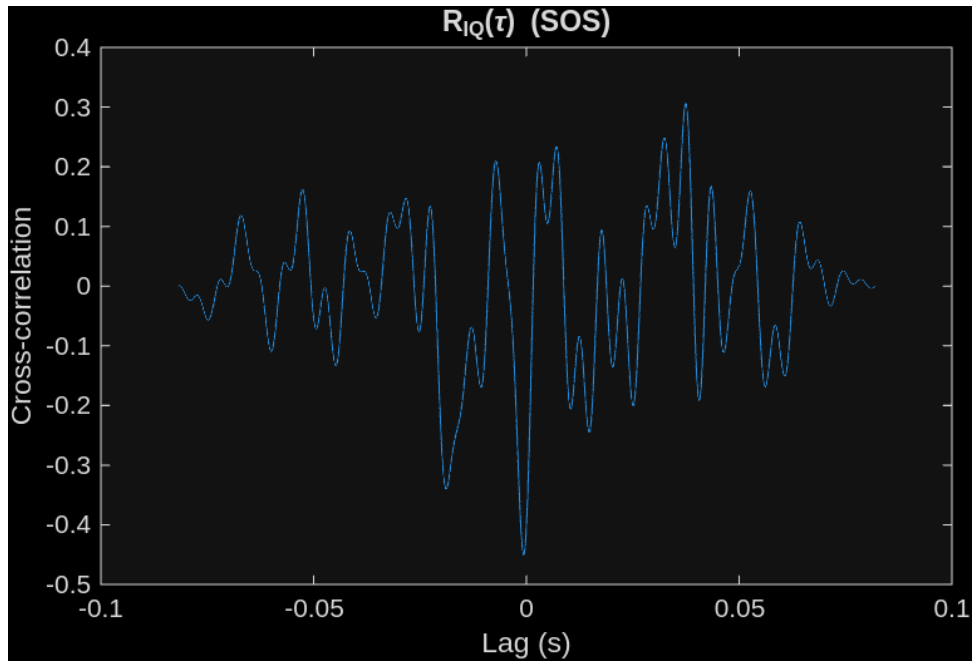


Figure 11: Cross-correlation (SOS model).

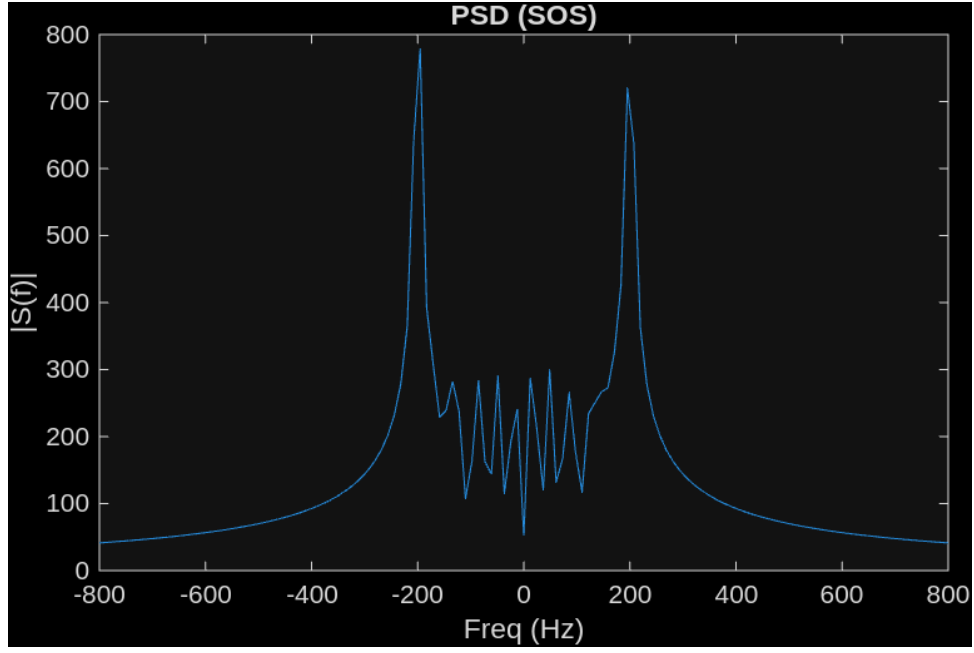


Figure 12: Power spectral density (SOS model).

1.5 D) Velocity = 3 m/s

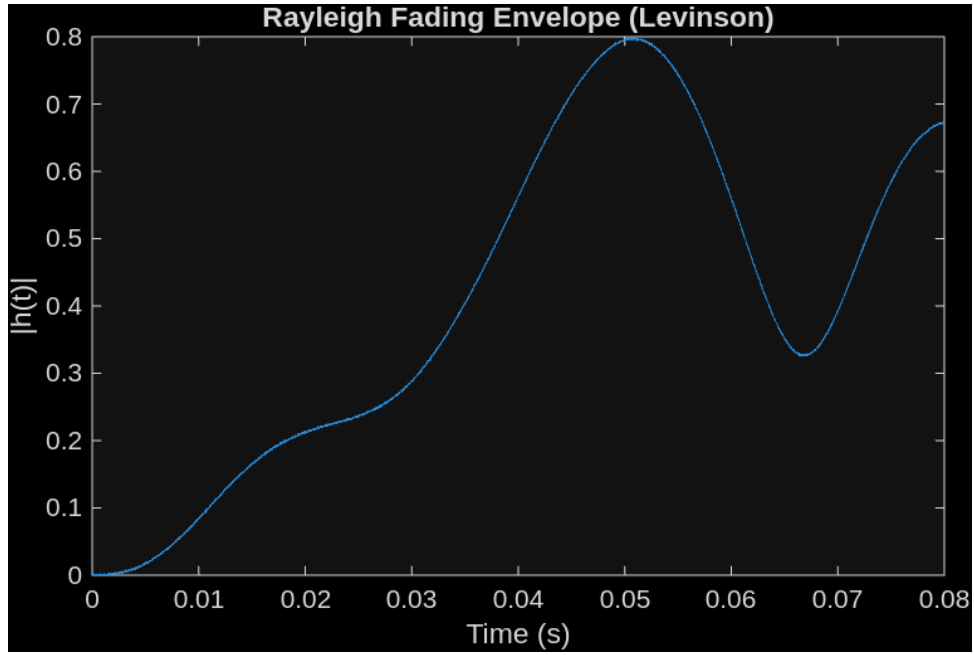


Figure 13: Path gain magnitude at $v = 3$ m/s (Levinson method).

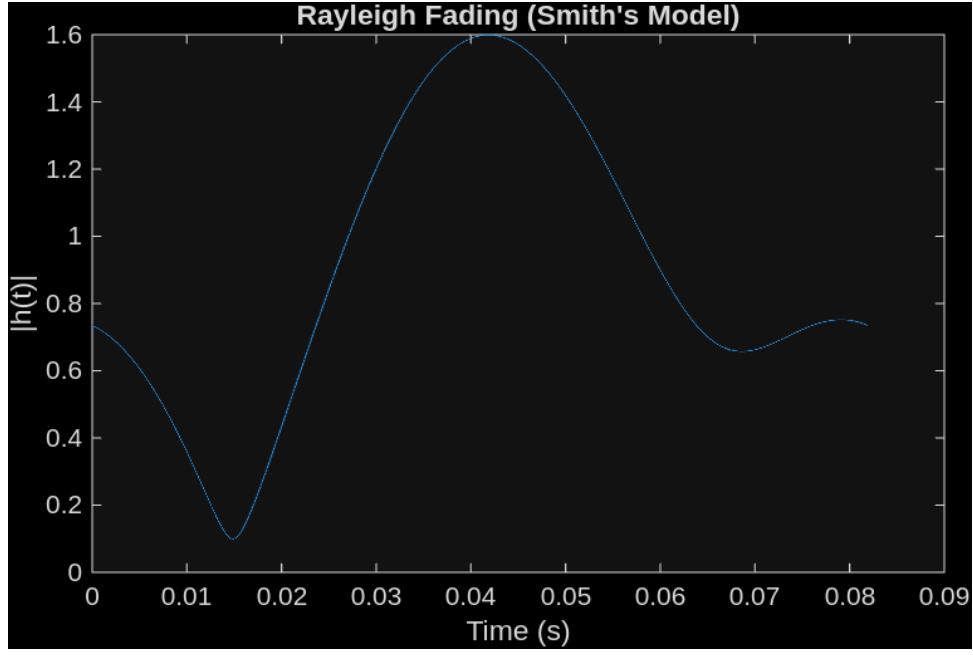


Figure 14: Path gain magnitude at $v = 3$ m/s (Smith's model).

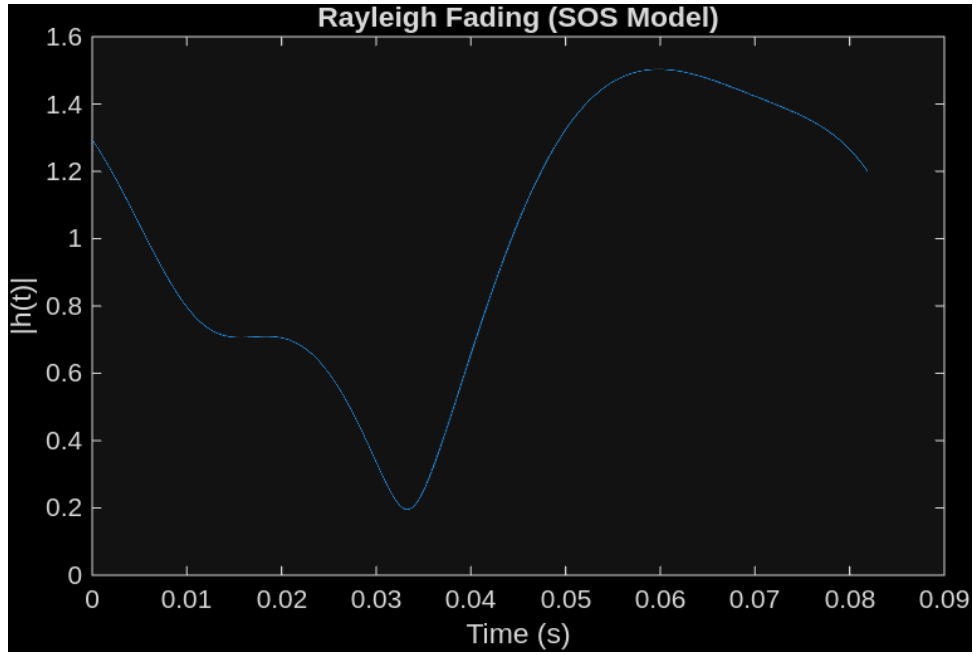


Figure 15: Path gain magnitude at $v = 3$ m/s (SOS model).

At $v = 3$ m/s the channel varies more slowly with time, leading to an increased coherence time.

2 Question 2

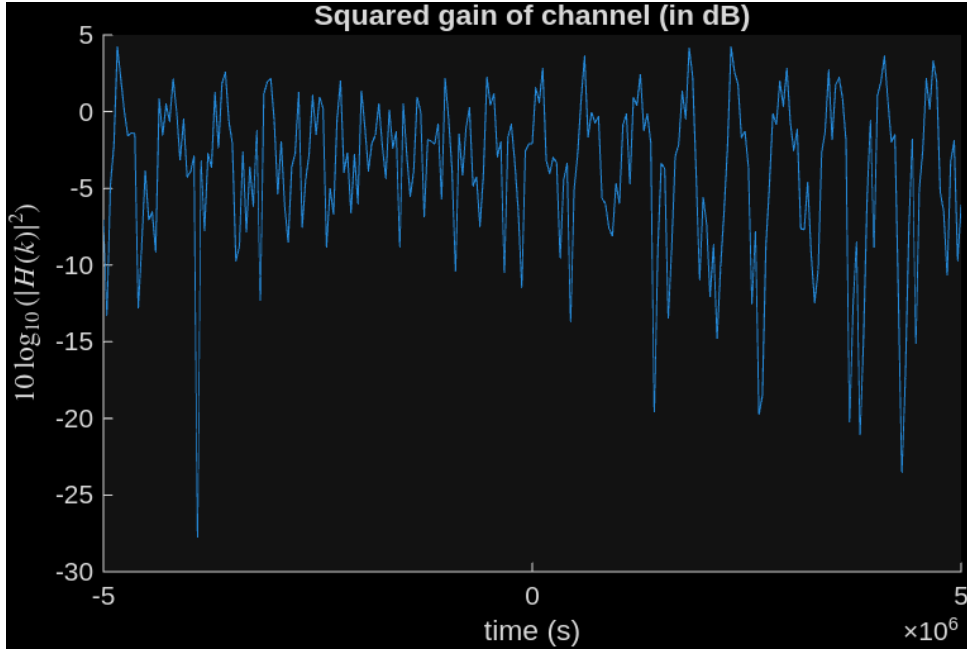


Figure 16: Variation of squared channel gain (dB) with frequency.

RMS Delay Spread

The mean excess delay is defined as:

$$\bar{\tau} = \sum_{i=0}^{L-1} p_i \tau_i \quad (13)$$

The RMS delay spread is:

$$\tau_{\text{rms}} = \sqrt{\sum_{i=0}^{L-1} p_i (\tau_i - \bar{\tau})^2} \quad (14)$$

The coherence bandwidth is inversely related to the RMS delay spread:

$$B_c \propto \frac{1}{\tau_{\text{rms}}} \quad (15)$$

For the given power delay profile (PDP), the RMS delay spread is calculated to be $\tau_{\text{rms}} = 2.33 \mu\text{s}$, which implies a coherence bandwidth on the order of 40 kHz. This is consistent with the frequency-selective behaviour observed in the plot.

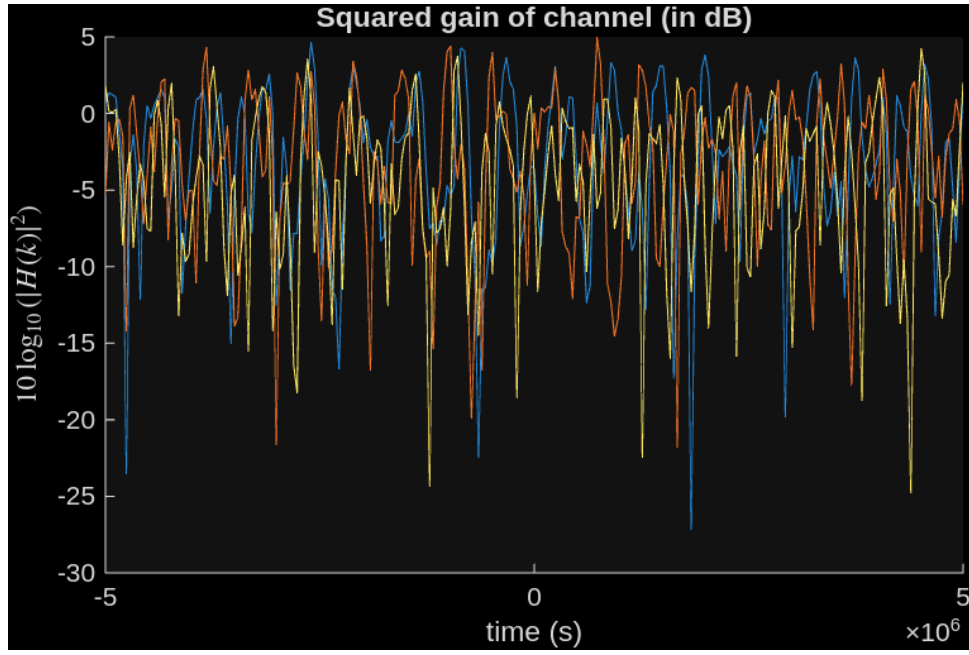


Figure 17: Combined plot of three independent channel realisations.

3 Question 3

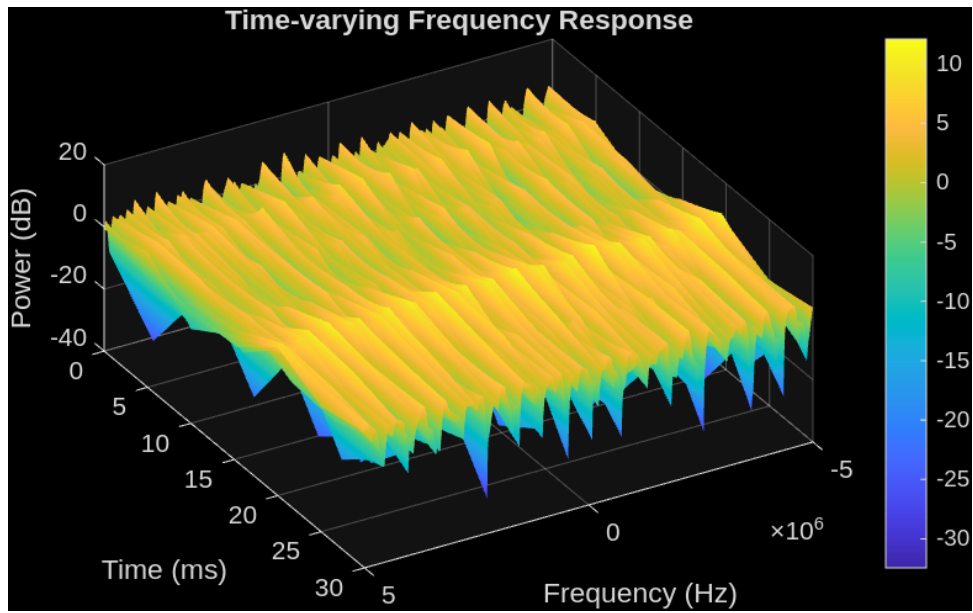


Figure 18: 3-D Doppler channel response.

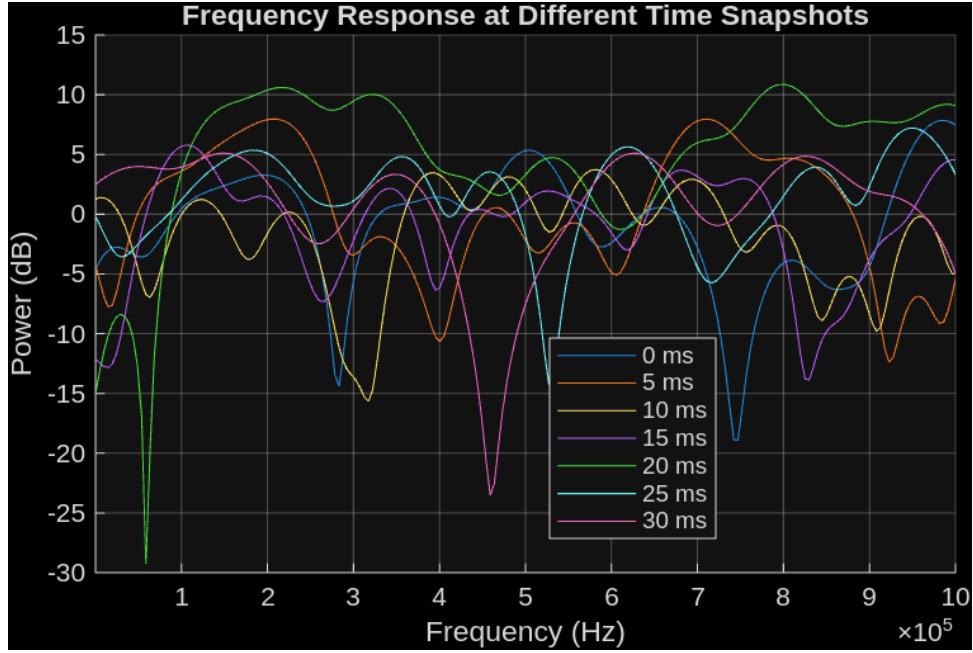


Figure 19: 2-D Doppler channel response.

The channel impulse response for the first time stamp is generated from the PDP in the same way as in Question 2. The channel is assumed to follow Clarke's model; subsequent time stamps are then generated using the Modified sum of sinusoids model applied independently to each delay index of the channel.

A MATLAB Code

A.1 Levinson Filter

Listing 1: Levinson noise-colouring filter.

```

1 function h = levinsonFilter(fd, Ts, p, N)
2     % --- Theoretical autocorrelation ---
3     m = 0:p;
4     R = besselj(0, 2*pi*fd*m*Ts); % Jakes autocorrelation
5     R(1) = R(1) + 1e-6;
6
7     % --- Levinson-Durbin ---
8     [a, E] = levinson(R, p); % a = [1 a1 ... ap]
9
10    % --- Generate white complex Gaussian ---
11    w = (randn(1, N) + 1j*randn(1, N))/sqrt(2);
12
13    % --- Generate coloured process ---
14    h = filter(sqrt(E), a, w);
15 end

```

A.2 Smith's Model

Listing 2: Smith's frequency-domain fading simulator.

```

1 function [h, t] = smithsFilter(fd, Ts, N)
2     df = 1/(N*Ts);
3     f = (-N/2 : N/2-1) * df;
4
5     % --- Jakes PSD ---
6     S = zeros(1, N);
7     idx = abs(f) <= fd;
8     S(idx) = 1 ./ sqrt(1 - (f(idx)/fd).^2 + eps); % avoid
          infinity
9
10    % --- Generate complex white Gaussian noise ---
11    W = (randn(1,N) + 1j*randn(1,N))/sqrt(2);
12
13    % --- Shape spectrum ---
14    H = sqrt(S) .* W;
15
16    % --- Time-domain fading process ---
17    h = ifft(ifftshift(H));
18
19    % --- Normalise power ---
20    h = h / sqrt(mean(abs(h).^2));
21
22    % --- Time axis ---
23    t = (0:N-1)*Ts;
24 end

```

A.3 Sum of Sinusoids Model

Listing 3: Modified sum-of-sinusoids fading model.

```

1 function [h, t] = SOS(fd, Ts, NO, N)
2     t = (0:N-1)*Ts;
3
4     % Angles
5     n = 1:NO;
6     theta = pi*(n - 0.5)/NO;
7
8     % Doppler frequencies
9     fn = fd * cos(theta);
10
11    % Random phases
12    phi = 2*pi*rand(1, NO);
13    psi = 2*pi*rand(1, NO);
14
15    % Generate fading
16    h = zeros(1, N);
17    for k = 1:NO
18        h = h + cos(2*pi*fn(k)*t + phi(k)) ...
19            + 1j*sin(2*pi*fn(k)*t + psi(k));

```

```
20     end
21
22     % Normalise
23     h = sqrt(2/N0) * h;
24 end
```